Aaron W. Storey

AI Research Engineer | Trustworthy AI Evaluation
Model Selection for Language-Based Decisions | Responsible AI
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Professional Summary

Research focus: deciding when a language classifier is adequate for a specific decision, turning the consequences of errors into measurable requirements, then selecting and qualifying models against them with explicit uncertainty. Also builds research prototypes coordinating models, tools, memory, and persistent state with evidence-tracing verification; applied evaluation spans biometrics and autonomous navigation. Brings 15 years of Army experience across medical operations, technical training, and information systems.

Technical Skills

Evaluation: Evaluation design, non-inferiority/equivalence testing, temperature/Platt calibration, selective prediction and deferral, Wilson/bootstrap intervals, ModernBERT/RoBERTa classifiers, LoRA fine-tuning, Hugging Face.
Agentic AI: Multi-agent systems, model orchestration, Model Context Protocol (MCP), tool use, agent memory, knowledge graphs, provenance, LLM evaluation.
ML and evaluation: Python, PyTorch, scikit-learn, Vision Transformers, face verification, face-image quality, calibration, uncertainty, explainable AI, reproducible experimentation.
Systems: SQL, R, Go, FastAPI, DuckDB, Docker, Linux, Git; ROS2, OpenCV, SLAM, state estimation, sensor fusion, GNSS-denied navigation.

Research and Engineering Experience

Independent AI Research Engineer Aug 2025 to Present
Kymera Systems LLC *San Antonio, TX*

- Develop agentic and multi-agent research prototypes combining heterogeneous models, structured tool use, and evaluation workflows.
- Design verification workflows that track intermediate artifacts, provenance, persistent state, and the evidence supporting generated conclusions; build experimental agent-memory systems as an individual contributor.

Graduate Research Assistant, Autonomous Systems and AI Jun 2026 to Sep 2026
Clarkson University, AVHBAC Laboratory *Remote*

- Developed simulation and evaluation workflows for GNSS-denied navigation, state estimation, sensor fusion, and lunar surface perception using Python, ROS2, and OpenCV.
- Investigated visual odometry, pose-graph optimization, and terrain-referenced localization for autonomous systems operating in physically constrained environments.
- Released **ofiqpy** on PyPI under AVHBAC, implementing the 27 BSI OFIQ v1.1.0 quality components plus its unified score.

Graduate Research Assistant, Trustworthy AI and Biometrics Jun 2025 to May 2026
Clarkson University, Center for Identification Technology Research (CITeR) *Remote*

- Conducted face-image-quality and biometric evaluation research involving ISO/IEC 29794-5, attribution faithfulness, calibration, uncertainty, and data quality.
- Built reproducible Python workflows for counterfactual analysis, perturbation testing, and statistical evaluation; contributed research manuscripts and protocols as a graduate researcher on DHS-supported CITeR work.

Education

Clarkson University Jan 2025 to Present
Ph.D. Candidate, Computer Science; GPA 3.98/4.00. Advisor: Jeanna Matthews. Member, Explainability & Transparency Group, since Sep 2026.

Maryville University of Saint Louis Aug 2023 to Dec 2024
M.S., Artificial Intelligence; GPA 3.97/4.00.

American Military University 2017
B.S., Information Technology; completed during Army service.

Selected Research Software

circle: Built a GitHub-native research collaboration framework in Go with shared memory, artifact tracking, review gates, a knowledge graph, and a 22-tool MCP server.

okf-tools: Built Python tools to validate and index Open Knowledge Format bundles, detecting broken relationships and semantic failures missed by basic conformance checks.

ofiqpy: Released the BSI OFIQ v1.1.0 quality-assessment profile in Python with configuration and model integrity checks and a C++ comparison workflow; PyPI v0.2.0, August 2026.

lunar-terrain-sidecar: Developed Godot terrain preprocessing using LOLA/PGDA products, ephemeris-derived illumination, and provenance controls. doi:10.5281/zenodo.22014046.

swinvit-face-verification: Built a common SwinFace/ArcFace evaluation interface using identical aligned inputs, checksum-verified models, and public LFW benchmark data.

medicaid-kg: Built a knowledge-graph and geospatial exploration system for 227 million HHS Medicaid provider-spending rows using DuckDB, NetworkX, FastAPI, React, and agent-assisted querying.

Selected Publications

Published. Kia, A.; **Storey, A. W.**; Imtiaz, M. *Advanced Hardware Security on Embedded Processors: A 2026 Systematic Review.* Electronics 15(5), 1135 (2026). doi:10.3390/electronics15051135.

Accepted and presented. Both AIBThings papers presented September 6, 2026; proceedings publication pending.

Storey, A. W.; Imtiaz, M. *A Three-Channel Audit of the Biometric Evidence Chain.* IEEE AIBThings 2026.

Storey, A. W.; Kia, A.; Imtiaz, M. *LLM on the Edge: A 2026 Perspective.* IEEE AIBThings 2026.

Accepted; presentation pending. Storey, A. W.; Ahmed, A.; Sukhdeve, Y.; Imtiaz, M. H.; Mondal, S. *Nonparametric Template Persistence Curves for Child Face Recognition via Beta Generalized Additive Models.* IEEE UEMCON 2026.

Earlier Technical and Operational Experience

United States Army Aug 2002 to Jan 2018

Defense Health Agency, San Antonio: Army assignments in data and operations analysis (Sep 2015 to Jan 2018) and educational technology (Jul 2013 to Sep 2015).

- Analyzed operational data, coordinated information-system projects, and developed medical training and simulation; evaluated 33 instructors during the Jun 2014 to May 2015 evaluation period.
- In Signal Plans and Operations, Kabul (2012 to 2013), advised military, civilian, and contractor personnel and led a U.S. systems team updating fiber-network infrastructure.
- Maintained accountability for equipment valued at over $3.2 million with zero loss during the Jul 2008 to Jun 2009 evaluation period, alongside medical operations and personnel supervision.

Career transition, Jan 2018 to Jul 2023: Personal responsibilities and self-directed technical study preceding graduate education in artificial intelligence.

Teaching, Service, and Credentials

Clarkson University: Nine guest lectures and course development for EE622 Advanced Biometrics (Summer 2025); invited LLM seminar (Jan 2026); three-session *Navigating Worlds Without GPS* series (Jun to Jul 2026).
100 Days of Machine Learning: Co-created a free, 35-lesson curriculum (Dec 2023 to Jan 2024).
CompTIA Security+ (earned during military service); IEEE Member since 2024; reviewer, IEEE AIBThings 2026.